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Thermodynamic assessment of the RE–Zn (RE = Dy, Er, Ho, Tb) binaries as a starting step for a Mg–Zn–Zr–Rare earth multicomponent database

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ABSTRACT

The study of Mg alloys containing Zn and Rare Earth (RE) elements is extremely important for several applications, such as bioimplants, due to their excellent biocompatibility and biocorrosion properties. Despite this fact, RE–Zn binary phase diagrams are not very well known, even though they are very similar to each other, being composed of several intermetallic compounds, usually concentrated near the Zn-rich side. In the present work, we propose a new thermodynamic assessment of four important RE–Zn (RE = Dy, Er, Ho, Tb) binary systems, following the CALPHAD method, as a launching board to the thermodynamic description of the Mg–Zn–Zr–RE multicomponent system. The liquid and bcc phases were described as substitutional solutions and all solid compounds were described as stoichiometric phases, thus avoiding models not easily extended to multicomponent systems and lacking implementation in widespread thermodynamic software. The resulting phase diagrams agree remarkably well with the available experimental data, making them ready to be incorporated into larger databases.

1. Introduction

The combination of Magnesium, Zinc, Zirconium and Rare Earth (RE) elements results in alloys that have received considerable attention due to their excellent mechanical and biocorrosion properties [1–3]. Consequently, phase relationships under both equilibrium and non-equilibrium conditions are extremely valuable for the design of alloy compositions and processing routes [4]. For that reason, it is important to characterize thermodynamically the phase diagrams of the binary and ternary subsystems involved.

With that in mind, RE–Zn phase diagrams are very similar to each other and slight differences in their topologies are related to the changes in the relative stabilities of the various binary intermediate phases [5,6]. Several RE–Zn phases diagrams obtained experimentally or calculated following one of several flavors of the CALPHAD approach [7] are reported in the literature: La–Zn [8–10], Ce–Zn [10–12], Pr–Zn [10,12], Gd–Zn [13,14], Nd–Zn [10,15,16], Sm–Zn [10,15,16], and Y–Zn [11,14,17].

The study of the RE–Zn systems using the CALPHAD method is important to increase the reliability of Mg–Zn–Zr–RE-based databases, for which there is very few experimental information available, but showing a promising range of biomedical applications [18]. In this work, thermodynamic reassessments of the Dy–Zn, Er–Zn, Ho–Zn and Tb–Zn binary systems were obtained through the CALPHAD [7] approach using the available experimental data in the literature. As a result, a more precise database, regarding especially the temperature of the invariant reactions, was obtained. In addition, all phase models are within the Compound Energy Formalism (CEF) [19], ready to be incorporated into multicomponent databases readable by thermodynamic software such as Thermo-Calc and Pandat [20].

2. Data on RE–Zn (RE = Dy, Er, Ho, Tb) systems

The Dy–Zn, Er–Zn, Ho–Zn and Tb–Zn systems were experimentally evaluated by Refs. [5,21] using differential thermal analysis (DTA), metallographic analysis, X-ray powder diffraction and electron probe microanalysis (EPMA). Zhu and Pelton [10,14] performed the thermodynamic assessments for these four RE–Zn systems using the modified quasi-chemical model (MQM) [16] for the liquid phase. In the rest of the section, we describe the expected equilibrium solid phases in these four RE–Zn phase diagrams and the overall characteristics of each system.

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Table 1
Crystallographic data for the stable solid phases in the Dy–Zn and Tb–Zn systems [5,21].

Phase	Pearson symbol	Space group	Prototype
α(Dy); α(Tb)	*hP*2	$P6_3/mmc$	Mg
β(Dy); β(Tb)	*cI*2	$Im\bar{3}m$	W
DyZn; TbZn	*cP*2	$Pm\bar{3}m$	CsCl
$DyZn_2$; $TbZn_2$	*oI*12	*Imma*	$CeCu_2$
$DyZn_3$; $TbZn_3$	*oP*16	*Pnma*	YZn_3
Dy_3Zn_{11}; Tb_3Zn_{11}	*oI*28	*Immm*	La_3Al_{11}
$Dy_{13}Zn_{58}$; $Tb_{13}Zn_{58}$	*hP*142	$P6_3/mmc$	$Gd_{13}Zn_{58}$
Dy_2Zn_{17}LT; Tb_2Zn_{17}LT	*hP*38	$P6_3/mmc$	Th_2Ni_{17}
Dy_2Zn_{17}HT; Tb_2Zn_{17}HT	*hR*57	$R\bar{3}m$	Th_2Zn_{17}
$DyZn_{12}$; $TbZn_{12}$	*tI*26	$I4/mmm$	$ThMn_{12}$
(Zn)	*hP*2	$P6_3/mmc$	Mg

2.1. Dy–Zn system

Seven intermetallic phases were proposed: DyZn, $DyZn_2$, $DyZn_3$, Dy_3Zn_{11}, $Dy_{13}Zn_{58}$, Dy_2Zn_{17} (LT and HT forms) and $DyZn_{12}$. The DyZn, $DyZn_2$, $Dy_{13}Zn_{58}$ and Dy_2Zn_{17} phases melt congruently at 1095 °C, 1050 °C, 930 °C and 930 °C, respectively, while $DyZn_3$, Dy_3Zn_{11} and $DyZn_2$ phases melt peritectically at 895 °C, 900 °C and 685 °C, respectively [21]. Zhu and Pelton [14] considered only the phases found by Saccone et al. [21] in their thermodynamic assessment of this system, even though the literature reports the $DyZn_5$ [22] and Dy_6Zn_{23} phases as well [23]. Crystallographic data for all solid phases of the Dy–Zn system are listed in Table 1 [21].

2.2. Tb–Zn system

Seven intermetallic phases were identified by Saccone et al. [5]: TbZn, $TbZn_2$, $TbZn_3$, Tb_3Zn_{11}, $Tb_{13}Zn_{58}$, Tb_2Zn_{17} (LT and HT forms) and $TbZn_{12}$. TbZn, $TbZn_2$, $Tb_{13}Zn_{58}$ and Tb_2Zn_{17} melt congruently at 1045 °C, 1028 °C, 907 °C and 915 °C, respectively. $TbZn_3$, Tb_3Zn_{11} and $TbZn_{12}$ melt peritectically at 900 °C, 895 °C and 673 °C, respectively. Zhu and Pelton [14], in the thermodynamic assessment of this system, considered only the phases found by Saccone et al. [5]. The polymorphic tranformation Tb_2Zn_{17}LT ↔ Tb_2Zn_{17}HT was considered in the present assessment using the same temperature proposed by Zhu and Pelton [14]. Crystallographic data for all solid phases of the Tb–Zn system are also listed in Table 1 [5].

2.3. Er–Zn system

Seven intermetallic phases were identified by Saccone et al. [5]: $ErZn_2$, $ErZn_3$, $Er_{13}Zn_{58}$, $ErZn_5$, Er_2Zn_{17} (LT and HT forms) and $ErZn_{12}$. The ErZn, $ErZn_2$, $Er_{13}Zn_{58}$ and Er_2Zn_{17} phases were reported to melt congruently at 1125 °C, 1065 °C, 940 °C and 890 °C respectively, while $ErZn_3$, $ErZn_5$ and $ErZn_{12}$ were reported to melt peritectically at 890 °C, ≈860 °C and 685 °C, respectively. As in the Dy–Zn system, Zhu and Pelton14] considered only the phases found by Saccone et al. [21] in their thermodynamic assessment. The polymorphic tranformation Er_2Zn_{17}LT ↔ Er_2Zn_{17}HT was considered in the present assessment using the same temperature proposed by Zhu and Pelton [14]. Crystallographic data for all solid phases of the Er–Zn system are listed in Table 2 [5].

2.4. Ho–Zn system

According to Saccone et al. [5], seven intermetallic phases exist in the Ho–Zn system: $HoZn_2$, $HoZn_3$, $Ho_{13}Zn_{58}$, $HoZn_5$, Ho_2Zn_{17} (LT and HT forms) and $HoZn_{12}$. HoZn, $HoZn_2$, $Ho_{13}Zn_{58}$ and Ho_2Zn_{17} melt congruently at 1109 °C, 1054 °C, 924 °C and 902 °C, respectively, while $HoZn_3$, $HoZn_5$ and $HoZn_{12}$ were reported to melt peritectically at 910 °C, 894 °C and 686 °C, respectively. Following the same procedure, the assessment by Zhu and Pelton [14] considered only the phases found by Saccone et al. [21]. The polymorphic transformation Ho_2Zn_{17}LT ↔ Ho_2Zn_{17}HT was considered in the present assessment using the same temperature proposed by Zhu and Pelton [14]. Crystallographic data for all solid phases of the Ho–Zn system are listed in Table 2 [5].

Table 2
Crystallographic data for the stable solid phases in the Er–Zn and Ho–Zn systems [5].

Phases	Pearson symbol	Space group	Prototype
(Er); (Ho)	*hP*2	$P6_3/mmc$	Mg
ErZn; HoZn	*cP*2	$Pm\bar{3}m$	CsCl
$ErZn_2$; $HoZn_2$	*oI*12	*Imma*	$CeCu_2$
$ErZn_3$; $HoZn_3$	*oP*16	*Pnma*	YZn_3
$Er_{13}Zn_{58}$; $Ho_{13}Zn_{58}$	hP142	$P6_3/mmc$	$Gd_{13}Zn_{58}$
$ErZn_5$; $HoZn_5$	*hP*36	$Pm\bar{3}m$	$ErZn_5$
Er_2Zn_{17}LT; Ho_2Zn_{17}LT	*hP*38	$P6_3/mmc$	Th_2Ni_{17}
Er_2Zn_{17}HT; Ho_2Zn_{17}HT	*hR*57	$R\bar{3}m$	Th_2Zn_{17}
$ErZn_{12}$; $HoZn_{12}$	*tI*26	$I4/mmm$	$ThMn_{12}$
(Zn)	*hP*2	$P6_3/mmc$	Mg

3. Phase models

The thermodynamic models adopted for the description of all phases follow the Compound Energy Formalism (CEF), which uses the concept of sublattices to describe each phase [19]. However, since none of the intermetallic phases show an appreciable experimental evidence for solubility, it is enough to describe here the formalism for stoichiometric and solution phases. All models for in the Dy–Zn, Er–Zn, Ho–Zn and Tb–Zn systems are given in Table 3, together with the values for the model parameters, to be briefly described in the following.

According to the CEF single-sublattice model, or substitutional model for short, the molar Gibbs energy of a (non-magnetic) phase φ is given by

$$G^{\varphi} = {}^{ref}G^{\varphi} + {}^{id}G^{\varphi} + {}^{ex}G^{\varphi} \tag{1}$$

where ${}^{ref}G^{\varphi}$ is the Gibbs energy of the mechanical mixture of the pure elements in phase φ referred to some reference state:

$${}^{ref}G^{\varphi} = \sum_i x_i {}^{o}G_i^{\varphi} \tag{2}$$

in which x_i is the composition of the element i, (i= {RE, Zn}) in the phase and ${}^{o}G_i^{\varphi}$ is the reference state for pure element i in phase φ. For the present work, we adopted the Standard Element Reference (SER) state as proposed by Dinsdale [24], with values taken from the SGTE unary (PURE5) database. ${}^{id}G^{\varphi}$ in Eq. (1) represents an ideal solution contribution to the Gibbs energy of mixing, considering that the atoms have no preferential neighbors in the solution. It is given by

$${}^{id}G^{\varphi} = RT\sum_i x_i \ln x_i \tag{3}$$

finally, ${}^{ex}G^{\varphi}$ is the excess term, relevant only to phases presenting solubility, in which case it is given by

$${}^{ex}G^{\varphi} = x_{Zn}x_{RE}\sum_{\nu=0}^{\nu_{max}} L_{Zn,RE}^{(\nu)}(x_{Zn} - x_{RE})^{\nu} \tag{4}$$

the excess parameters $L_{Zn,RE}^{(\nu)}$ are usually expanded in a linear, temperature-only dependent form $L_{Zn,RE}^{(\nu)} = a_{\nu} + b_{\nu}T$. In the present work, the excess term is only used for the liquid and β (bcc) phases, since all remaining solid phases show no appreciable solubility.

Table 3
Optimized parameters and models for the stable phases in the Dy–Zn, Er–Zn, Ho–Zn and Tb–Zn phase diagrams (in J/mol of atoms). ${}^{o}G^{SER}_{Dy}$, ${}^{o}G^{SER}_{Er}$, ${}^{o}G^{SER}_{Ho}$, ${}^{o}G^{SER}_{Tb}$ and ${}^{o}G^{SER}_{Zn}$ are the Standard Element Reference state for Dy, Er, Ho, Tb and Zn, respectively, taken from the PURE5 database [24].

Phase	CEF model	Thermodynamic parameters (J/mol of atoms)
Liquid	(Dy,Zn)	$L^{0}_{Dy,Zn} = -103075 + 6.233T$
		$L^{1}_{Dy,Zn} = +57847 - 10.801T$
β(Dy)	$(Dy,Zn)_1(VA)_3$	$L^{0}_{\beta(Dy)} = -45300 - 3.000T$
DyZn	$(Dy)_1(Zn)_1$	$G^{DyZn} = -67640 + 0.100T + 1{}^{o}G^{SER}_{Dy} + 1{}^{o}G^{SER}_{Zn}$
$DyZn_2$	$(Dy)_1(Zn)_2$	$G^{DyZn_2} = -110592 + 4.770T + 1{}^{o}G^{SER}_{Dy} + 2{}^{o}G^{SER}_{Zn}$
$DyZn_3$	$(Dy)_1(Zn)_3$	$G^{DyZn_3} = -139800 + 14.660T + 1{}^{o}G^{SER}_{Dy} + 3{}^{o}G^{SER}_{Zn}$
Dy_3Zn_{11}	$(Dy)_3(Zn)_{11}$	$G^{Dy_3Zn_{11}} = -455056 + 44.500T + 3{}^{o}G^{SER}_{Dy} + 11{}^{o}G^{SER}_{Zn}$
$Dy_{13}Zn_{58}$	$(Dy)_{13}(Zn)_{58}$	$G^{Dy_{13}Zn_{58}} = -2153856 + 195.300T + 13{}^{o}G^{SER}_{Dy} + 58{}^{o}G^{SER}_{Zn}$
Dy_2Zn_{17}LT	$(Dy)_2(Zn)_{17}$	$G^{Dy_2Zn_{17}LT} = -419520 + 32.750T + 2{}^{o}G^{SER}_{Dy} + 17{}^{o}G^{SER}_{Zn}$
Dy_2Zn..HT	$(Dy)_2(Zn)_{17}$	$G^{Dy_2Zn_{17}HT} = -400000 + 14.080T + 2{}^{o}G^{SER}_{Dy} + 17{}^{o}G^{SER}_{Zn}$
$DyZn_{12}$	$(Dy)_1(Zn)_{12}$	$G^{DyZn_{12}} = -219500 + 15.852T + 1{}^{o}G^{SER}_{Dy} + 12{}^{o}G^{SER}_{Zn}$
Liquid	(Er,Zn)	$L^{0}_{Er,Zn} = -103053 + 5.905T$
		$L^{1}_{Er,Zn} = +57888 - 20.278T$
ErZn	$(Er)_1(Zn)_1$	$G^{ErZn} = -66432 + 0.007T + 1{}^{o}G^{SER}_{Dy} + 1{}^{o}G^{SER}_{Zn}$
$ErZn_2$	$(Er)_1(Zn)_2$	$G^{ErZn_2} = -111456 + 9.023T + 1{}^{o}G^{SER}_{Er} + 2{}^{o}G^{SER}_{Zn}$
$ErZn_3$	$(Er)_1(Zn)_3$	$G^{ErZn_3} = -137000 + 17.200T + 1{}^{o}G^{SER}_{Er} + 3{}^{o}G^{SER}_{Zn}$
$Er_{13}Zn_{58}$	$(Er)_{13}(Zn)_{58}$	$G^{Er_{13}Zn_{58}} = -2156696 + 288.000T + 13{}^{o}G^{SER}_{Er} + 58{}^{o}G^{SER}_{Zn}$
$ErZn_5$	$(Er)_1(Zn)_5$	$G^{ErZn_5} = -174864 + 25.700T + 1{}^{o}G^{SER}_{Er} + 5{}^{o}G^{SER}_{Zn}$
Er_2Zn_{17}LT	$(Er)_2(Zn)_{17}$	$G^{Er_2Zn_{17}LT} = -395808 + 39.330T + 2{}^{o}G^{SER}_{Er} + 17{}^{o}G^{SER}_{Zn}$
Er_2Zn_{17}HT	$(Er)_2(Zn)_{17}$	$G^{Er_2Zn_{17}HT} = -380000 + 23.900T + 2{}^{o}G^{SER}_{Er} + 17{}^{o}G^{SER}_{Zn}$
$ErZn_{12}$	$(Er)_1(Zn)_{12}$	$G^{ErZn_{12}} = -198432 + 9.548T + 1{}^{o}G^{SER}_{Er} + 12{}^{o}G^{SER}_{Zn}$
Liquid	(Ho,Zn)	$L^{0}_{Ho,Zn} = -103056 + 1.591T$
		$L^{1}_{Ho,Zn} = +57834 - 15.347T$
HoZn	$(Ho)_1(Zn)_1$	$G^{HoZn} = -71412 + 0.900T + 1{}^{o}G^{SER}_{Dy} + 1{}^{o}G^{SER}_{Zn}$
$HoZn_2$	$(Ho)_1(Zn)_2$	$G^{HoZn_2} = -111744 + 4.170T + 1{}^{o}G^{SER}_{Ho} + 2{}^{o}G^{SER}_{Zn}$
$HoZn_3$	$(Ho)_1(Zn)_3$	$G^{HoZn_3} = -137500 + 11.730T + 1{}^{o}G^{SER}_{Ho} + 3{}^{o}G^{SER}_{Zn}$
$Ho_{13}Zn_{58}$	$(Ho)_{13}(Zn)_{58}$	$G^{Ho_{13}Zn_{58}} = -2122900 + 170.000T + 13{}^{o}G^{SER}_{Ho} + 58{}^{o}G^{SER}_{Zn}$
$HoZn_5$	$(Ho)_1(Zn)_5$	$G^{HoZn_5} = -172800 + 16.500T + 1{}^{o}G^{SER}_{Ho} + 5{}^{o}G^{SER}_{Zn}$
Ho_2Zn_{17}LT	$(Ho)_2(Zn)_{17}$	$G^{Ho_2Zn_{17}LT} = -420000 + 38.140T + 2{}^{o}G^{SER}_{Ho} + 17{}^{o}G^{SER}_{Zn}$
Ho_2Zn_{17}HT	$(Ho)_2(Zn)_{17}$	$G^{Ho_2Zn_{17}HT} = -408576 + 27.000T + 2{}^{o}G^{SER}_{Ho} + 17{}^{o}G^{SER}_{Zn}$
$HoZn_{12}$	$(Ho)_1(Zn)_{12}$	$G^{HoZn_{12}} = -221936 + 20.545T + 1{}^{o}G^{SER}_{Ho} + 12{}^{o}G^{SER}_{Zn}$
Liquid	(Tb,Zn)	$L^{0}_{Tb,Zn} = -103063 - 3.502T$
		$L^{1}_{Tb,Zn} = +57846 - 6.538T$
β(Tb)	$(Tb,Zn)_1(VA)_3$	$L^{0}_{\beta(Dy)} = -61400$
TbZn	$(Tb)_1(Zn)_1$	$G^{TbZn} = -73204 + 0.050T + 1{}^{o}G^{SER}_{Tb} + 1{}^{o}G^{SER}_{Zn}$
$TbZn_2$	$(Tb)_1(Zn)_2$	$G^{TbZn_2} = -113842 + 0.200T + 1{}^{o}G^{SER}_{Tb} + 2{}^{o}G^{SER}_{Zn}$
$TbZn_3$	$(Tb)_1(Zn)_3$	$G^{TbZn_3} = -148000 + 12.900T + 1{}^{o}G^{SER}_{Tb} + 3{}^{o}G^{SER}_{Zn}$
Tb_3Zn_{11}	$(Tb)_3(Zn)_{11}$	$G^{Tb_3Zn_{11}} = -472000 + 32.000T + 3{}^{o}G^{SER}_{Tb} + 11{}^{o}G^{SER}_{Zn}$
$Tb_{13}Zn_{58}$	$(Tb)_{13}(Zn)_{58}$	$G^{Tb_{13}Zn_{58}} = -2187936 + 106.500T + 13{}^{o}G^{SER}_{Tb} + 58{}^{o}G^{SER}_{Zn}$
Tb_2Zn_{17}LT	$(Tb)_2(Zn)_{17}$	$G^{Tb_2Zn_{17}LT} = -432288 + 20.320T + 2{}^{o}G^{SER}_{Tb} + 17{}^{o}G^{SER}_{Zn}$
Tb_2Zn_{17}HT	$(Tb)_2(Zn)_{17}$	$G^{Tb_2Zn_{17}HT} = -428000 + 16.138T + 2{}^{o}G^{SER}_{Tb} + 17{}^{o}G^{SER}_{Zn}$
$TbZn_{12}$	$(Tb)_1(Zn)_{12}$	$G^{TbZn_{12}} = -216511 + 0.100T + 1{}^{o}G^{SER}_{Tb} + 12{}^{o}G^{SER}_{Zn}$

3.1. Stoichiometric phases

The Dy–Zn, Er–Zn, Ho–Zn and Tb–Zn systems have seven stoichiometric phases each [21,5], namely, DyZn, $DyZn_2$, $DyZn_3$, Dy_3Zn_{11}, $Dy_{13}Zn_{58}$, Dy_2Zn_{17}, $DyZn_{12}$, ErZn, $ErZn_2ErZn_3$, $Er_{13}Zn_{58}$, $ErZn_5$, Er_2Zn_{17}, $ErZn_{12}$, HoZn, $HoZn_2$, $HoZn_3$, $Ho_{13}Zn_{58}$, $HoZn_5$, Ho_2Zn_{17}, $HoZn_{12}$, TbZn, $TbZn_2$, $TbZn_3$, Tb_3Zn_{11}, $Tb_{13}Zn_{58}$, Tb_2Zn_{17}, $TbZn_{12}$. These phases are represented as

$$(A_1)_{a1}: (A_2)_{a2}: (A_3)_{a3}: \ldots,$$

in which a_i indicates the stoichiometry of each sublattice, singly occupied by component A_i. Since there is no solubility of any element, the ideal term vanishes in Eq. (1) and the reference term can be rewritten as

$${}^{ref}G^{\varphi} = \sum_i a_i\,{}^{o}G_i + \Delta_f G^{\varphi} \tag{5}$$

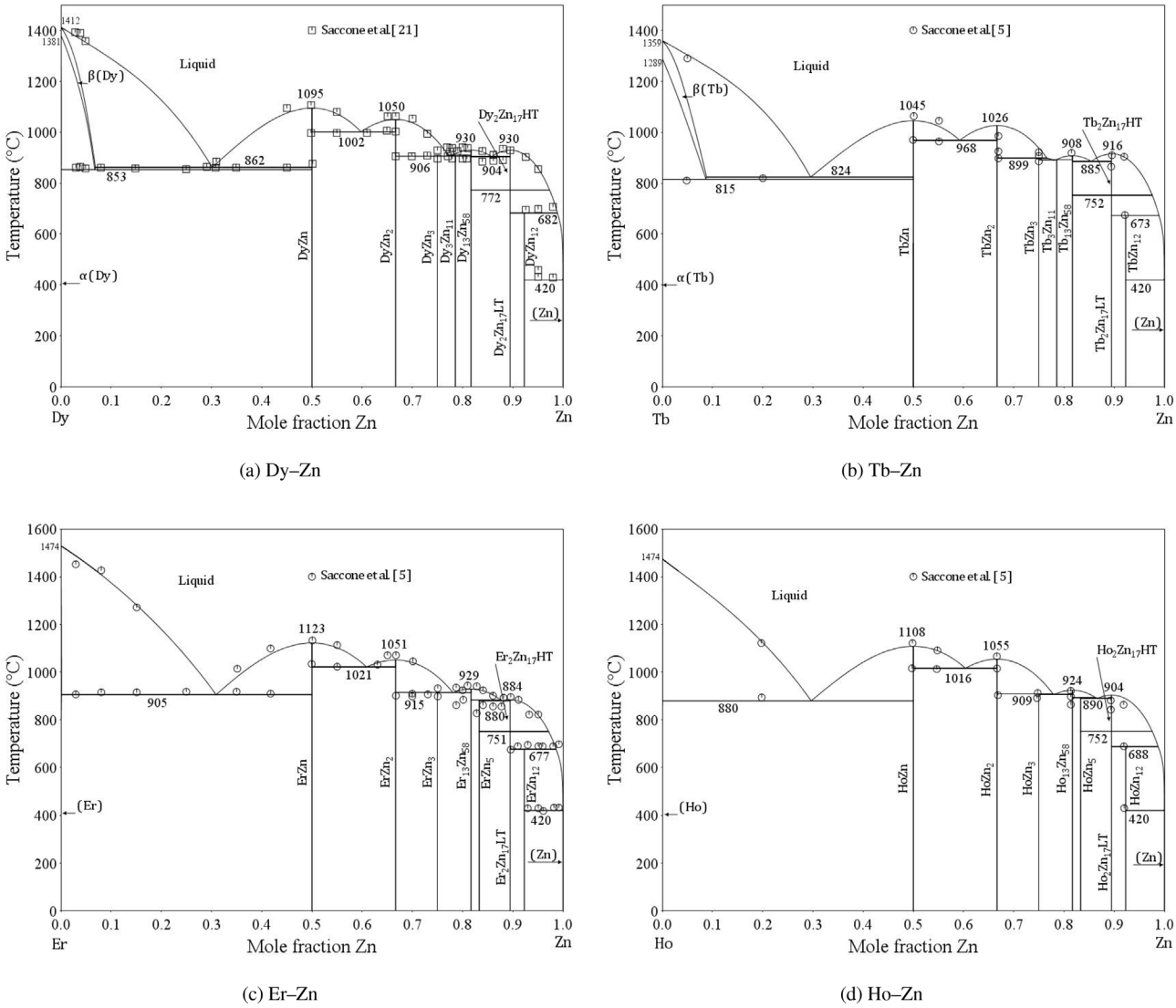

Fig. 1. Calculated (a) Dy–Zn, (b) Tb–Zn, (c) Er–Zn, and (d) Ho–Zn phase diagrams, in comparison to selected experimental data [5,21].

where $i = \{RE, \ Zn\}$, ${}^{o}G_i$ is the partial Gibbs free energy of element i in the Standard Element Reference (SER) state, and $\Delta^f G^{\varphi}$ is the molar Gibbs energy of formation of the compound φ (referred to the pure elements in the SER). It is usually written as a linear polynomial in temperature:

$$\Delta_f G^{\varphi} = A^{\varphi} - B^{\varphi} T \tag{6}$$

with A_{φ} and B_{φ} being generally called enthalpic and entropic contributions, respectively.

4. Optimization procedure

The thermodynamic assessment was carried out using the PARROT module [25] of the Thermo-Calc computational package [26], following well-established procedures [27]. As mentioned previously, all pure element data were taken from the PURE5 database [24]. During the optimization of the model parameters, according to the reliability of each experiment, different weights were set in order to get a better overall description, respecting all the available data in the literature. Since experiments on the liquid phase tend to be crucial to a thermodynamic optimization, leading to a more reliable description, higher weights were attributed to the invariant reactions involving the liquid phase. Care was taken to assure that stable phases at room temperature remain stable also at lower temperatures, i.e., no phase transitions occur below 298.15 K. In order to achieve this, some ab-initio values for the formation energies at 0 K were taken from the Materials Project website [28], with some adjustments to guarantee a good description at all temperatures. Besides, we detected a problem with the PURE5 description of the liquid phase for pure Tb: it becomes stable at $T \lesssim 50$K, affecting also the binary Tb–Zn phase diagram. Such problem is not present in the case of the other RE elements. To circumvent this issue, we added a "patch" to the pure Tb liquid parameter in the Tb–Zn database so that the reappearance of the liquid at very low temperatures is avoided. Despite all care taken to improve the region below room temperature, we highlight the fact that our database is strictly valid for $T > 298.15$K, as no data below that limit (besides the ab initio values) was taken into consideration during the optimization.

5. Results and discussion

The resulting phase diagrams, according to the thermodynamic optimization procedure described above and using the models and parameters given in Table 3, are shown in Fig. 1a–d. All four phase diagrams are very similar to each other but, for the purposes of the following discussion, the four RE–Zn phase diagrams are separated in two groups: the Dy–Zn/Tb–Zn systems, in which we expect the

Table 4
Equilibrium invariant reactions in the stable Dy–Zn and Tb –Zn binary systems. Calculated temperatures and liquid compositions at invariant reactions are shown in comparison to experimental values by Saccone et al. [5,21] and calculated by Zhu and Pelton [14].

		Temperature (°C)			x_{Zn}^{liq}		
Reaction	Type	Calc.	Exp [21].	[14]	Calc.	Exp [21].	[14]
Liq ↔ β(Dy) + DyZn	Eutectic	862	–	864	0.30	–	0.27
β(Dy) ↔α(Dy) + DyZn	Eutectoid	853	–	848	0.07	–	0.07
Liq ↔ DyZn	Congruent	1095	1095	1081	0.50	0.50	0.50
Liq ↔ DyZn + $DyZn_2$	Eutectic	1002	991	1003	0.60	0.61	0.60
Liq ↔ $DyZn_2$	Congruent	1050	1050	1037	0.66	0.66	0.66
Liq + $DyZn_2$ ↔ $DyZn_3$	Peritectic	906	895	895	0.77	0.75	0.76
Liq ↔ $DyZn_3$ + Dy_3Zn_{11}	Eutectic	905	885	887	0.77	0.76	0.76
Liq + $Dy_{13}Zn_{58}$ ↔ Dy_3Zn_{11}	Peritectic	907	900	891	0.78	0.77	0.77
Liq ↔ $Dy_{13}Zn_{58}$	Congruent	930	930	925	0.82	0.82	0.82
Liq ↔ $Dy_{13}Zn_{58}$ + Dy_2Zn_{17}	Eutectic	904	875	891	0.86	0.85	0.86
Liq ↔ Dy_2Zn_{17}HT	Congruent	930	930	914	0.90	0.90	0.90
$Dy_2Zn_{17}HT$ ↔ $Dy_2Zn_{17}LT$	Polimorfic	772	–	772	0.90	–	0.90
Liq + Dy_2Zn_{17} ↔ $DyZn_{12}$	Peritectic	682	685	677	0.99	0.98	0.99
Liq ↔ $DyZn_{12}$ + (Zn)	Eutectic	420	420	420	> 0.99	> 0.99	> 0.99
		Calc.	Exp [5].	[14]	Calc.	Exp [5].	[14]
Liq ↔ β(Tb) + TbZn	Eutectic	824	–	823	0.30	–	0.27
β(Tb) ↔α(Tb) + TbZn	Eutectoid	815	–	809	0.09	–	0.10
Liq ↔ TbZn	Congruent	1045	1045	1046	0.50	0.50	0.50
Liq ↔ TbZn + $TbZn_2$	Eutectic	968	960	976	0.59	0.60	0.60
Liq ↔ $TbZn_2$	Congruent	1026	1028	1016	0.66	0.66	0.66
Liq + $TbZn_2$ ↔ $TbZn_3$	Peritectic	899	900	889	0.77	0.76	0.75
Liq ↔ $TbZn_3$ + Tb_3Zn_{11}	Eutectic	891	881	886	0.78	0.77	0.76
Liq + $Tb_{13}Zn_{58}$ ↔ Tb_3Zn_{11}	Peritectic	892	895	890	0.78	0.79	0.78
Liq ↔ $Tb_{13}Zn_{58}$	Congruent	908	907	909	0.82	0.82	0.82
Liq ↔ $Tb_{13}Zn_{58}$ + Tb_2Zn_{17}	Eutectic	885	889	886	0.86	0.85	0.86
Liq ↔ $Tb_2Zn_{17}HT$	Congruent	916	915	905	0.90	0.90	0.90
$Tb_2Zn_{17}HT$ ↔ $Tb_2Zn_{17}LT$	Polimorfic	752	–	750	0.90	–	0.90
Liq + Tb_2Zn_{17} ↔ $TbZn_{12}$	Peritectic	673	673	668	0.99	0.99	0.99
Liq ↔ $TbZn_{12}$ + (Zn)	Eutectic	420	419.6	420	> 0.99	> 0.99	> 0.99

Table 5
Equilibrium invariant reactions in the stable Er–Zn and Ho–Zn binary systems. Calculated temperatures and liquid compositions at invariant reactions are shown in comparison to experimental values by Saccone et al. [5] and calculated by Zhu and Pelton [14].

		Temperature (°C)			x_{Zn}^{liq}		
Reaction	Type	Calc.	Exp [5].	[14]	Calc.	Exp [5].	[14]
Liq ↔ (Er) + ErZn	Eutectic	905	905	910	0.31	0.31	0.29
Liq ↔ ErZn	Congruent	1123	1125	1101	0.50	0.50	0.50
Liq ↔ ErZn + $ErZn_2$	Eutectic	1021	1015	1026	0.61	0.60	0.61
Liq ↔ $ErZn_2$	Congruent	1051	1065	1047	0.66	0.66	0.66
Liq + $ErZn_2$ ↔ $ErZn_3$	Peritectic	915	890	912	0.78	0.75	0.76
Liq ↔ $ErZn_3$ + $Er_{13}Zn_{58}$	Eutectic	914	850	889	0.78	0.77	0.77
Liq ↔ $Er_{13}Zn_{58}$	Congruent	929	940	928	0.82	0.82	0.82
Liq + $Er_{13}Zn_{58}$ ↔ $ErZn_5$	Peritectic	883	860	882	0.86	0.86	0.87
Liq ↔ $ErZn_5$ + Er_2Zn_{17}	Eutectic	880	845	874	0.87	0.87	0.87
Liq ↔ Er_2Zn_{17}	Congruent	884	890	890	0.90	0.90	0.90
Er_2Zn_{17}HT ↔ Er_2Zn_{17}LT	Polimorfic	751	–	750	0.90	–	0.90
Liq + Er_2Zn_{17} ↔ $ErZn_{12}$	Peritectic	677	685	673	0.99	0.99	0.99
Liq ↔ $ErZn_{12}$ + (Zn)	Eutectic	420	419.6	420	> 0.99	> 0.99	> 0.99
Liq ↔ (Ho) + HoZn	Eutectic	880	878	905	0.30	0.30	0.28
Liq ↔ HoZn	Congruent	1108	1109	1105	0.50	0.50	0.50
Liq ↔ HoZn + $HoZn_2$	Eutectic	1016	1000	1025	0.60	0.60	0.61
Liq ↔ $HoZn_2$	Congruent	1055	1054	1050	0.66	0.66	0.66
Liq + $HoZn_2$ ↔ $HoZn_3$	Peritectic	909	910	909	0.77	0.76	0.76
Liq ↔ $HoZn_3$ + $Ho_{13}Zn_{58}$	Eutectic	907	884	896	0.78	0.79	0.78
Liq ↔ $Ho_{13}Zn_{58}$	Congruent	924	924	914	0.82	0.82	0.82
Liq + $Ho_{13}Zn_{58}$ ↔ $HoZn_5$	Peritectic	893	894	874	0.85	0.84	0.86
Liq ↔ $HoZn_5$ + Ho_2Zn_{17}	Eutectic	890	855	868	0.87	0.86	0.87
Liq ↔ Ho_2Zn_{17}HT	Congruent	904	902	872	0.90	0.90	0.90
Ho_2Zn_{17}HT ↔ Ho_2Zn_{17}LT	Polimorfic	752	–	750	0.90	–	0.90
Liq + Ho_2Zn_{17} ↔ $HoZn_{12}$	Peritectic	688	686	681	0.99	0.99	0.99
Liq ↔ $HoZn_{12}$ + (Zn)	Eutectic	420	419.6	420	> 0.99	> 0.99	> 0.99

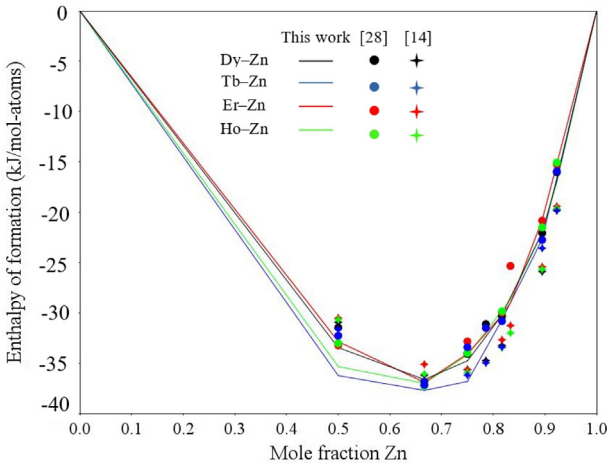

Fig. 2. Standard enthalpies of formation at 298.15 K for binary Dy–Zn, Er–Zn, Tb–Zn and Ho–Zn compounds, as determined in the present work, compared with the values given by Zhu and Pelton [14] and the ab initio values from the Materials Project [28].

appearance of the β (bcc) phases close to the RE-rich side, and the Er–Zn/Ho–Zn phase diagrams, in which the allotropic transformation of the RE element is not present.

The Dy–Zn and Tb–Zn phase diagrams are shown in Fig. 1a and b, respectively, together with experimental data for comparison [5,21]. A good agreement with the experimental data for both phase diagrams has been reached, as can be seen from the Figures, and also from Table 4, which shows the temperatures and liquid compositions of the invariant reactions, comparing the available experimental data to those given by the present assessment and a previous assessment [14]. All invariant reaction temperatures show a good agreement with the available experimental data. The temperatures of all invariant reactions agree satisfactorily with the experimental values It should be stressed, moreover, that some of the invariant temperatures were not experimentally measured by Saccone et al. [5,21], but were simply estimated based on the surrounding transformations. It should be noted, moreover, that the experimental interpretation of the invariant reactions involving the liquid phase close to the RE-rich corner, according to Saccone et al. [21], in the case of Dy–Zn, and Saccone et al. [5] for Tb–Zn, differ from the interpretation adopted here. In the present work, we followed the strategy adopted by Ref. [14]. It means that, according to the evidence on Gd–Zn [29] and the fact that all RE–Zn phase diagrams are very similar, we made the bcc phase stable down to an eutectoid reaction (β-RE → α-RE + REZn) close to the eutectic reaction L → β-RE + REZn. This differs from the proposed experimental phase diagrams [5,21], in which the bcc phase has no appreciable Zn solubility. The decision made in the present assessment to extend the bcc single phase field, thus, keeps our assessment in line with the recent assessments [10,14] of similar systems, but, in the present case, using only CEF phase models. Furthermore, in this work a better fit of the congruent melting temperature of the DyZn, $DyZn_2$, $Dy_{13}Zn_{58}$, Dy_2Zn_{17}, $TbZn_{12}$, TbZn, $TbZn_2$, $Tb_{13}Zn_{58}$ and Tb_2Zn_{17} stoichiometric phases were obtained when compared to experimental data [5,21], as can be seen in Table 4.

Fig. 1c and d show the phase diagrams according to the thermodynamic optimizations procedures of the Er–Zn and Ho–Zn systems, with experimental data from Saccone et al. [5] for comparison. As in the previous cases, the Er–Zn and Ho–Zn phase diagrams show a good agreement with the experimental data [5]. Table 5 shows the temperatures and liquid compositions of the invariant reactions, comparing the available experimental data to those given by the present assessment. As in the case of the binaries described in the preceding paragraph, all temperatures agree very well with the experimental values, except a few cases, mainly in the case of the Er–Zn system. The same observations made before, regarding some of the experiments being merely estimations, are also valid here. From Table 5, it can also be seen that, in this work, a better overall fit of the invariant reactions and congruent melting temperatures for the ErZn, $ErZn_2$, $Er_{13}Zn_{58}$, Er_2Zn_{17}, HoZn, $HoZn_2$, $Ho_{13}Zn_{58}$ and Ho_2Zn_{17} phases were obtained, compared to the values reached by the assessment proposed by Zhu and Pelton [14].

Finally, Fig. 2 shows a comparison between calculated standard enthalpies of formation at 298.15 K for the four RE–Zn systems investigated here with the results from Zhu and Pelton [14] and the ab initio results from the Materials Project [28]. As seen from Fig. 2, our results agree satisfactorily well with both data sets. We stress the fact that no experimental enthalpy data are available for the four RE-Zn systems discussed here. On the other hand, our values are compatible with the experimental data found in similar RE-Zn binary systems, such as La–Zn [30], Ce–Zn [31] and Nd–Zn [23], which show values within the same order of magnitude as the values shown in Fig. 2. Given the observed similarity among different RE elements with Zn, this agreement is a strong indication that our assessment is thermodynamically consistent.

6. Conclusions

In the present work, new thermodynamic assessments were proposed for four important rare earth (RE) binary systems with Zn: Dy–Zn, Er–Zn, Ho–Zn, and Tb–Zn. The results agree remarkably well with the experimental data found in the literature, regarding invariant reaction and congruent melting temperatures. Only phase models based on the Compound Energy Formalism (CEF) were employed, which makes the assessments readily applicable to incorporation into large multicomponent databases, to be used in virtually all available thermodynamic software. Following this strategy, we have an ongoing project to use the already assessed systems as a launching board for the development of a multipurpose, multicomponent Mg–Zn–Zr–RE thermodynamic database.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2019.101676.

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